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Capital Market Equilibrium without Riskless Assets: Heterogeneous Expectations^{*†}

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Abstract

The existence theorem of Allingham (1991) for the capital asset pricing model (CAPM) is generalized to the case where agents have heterogeneous expectations on the return distribution and the mean-variance utility functions are quasiconcave. The consequence of the paper is based upon new conditions which are quite different from and weaker than the literature. A new argument is required to cover the case of heterogeneous expectations.

KEYWORDS: Asset Market Equilibrium, Satiation, CAPM, Heterogeneous Expectations.

JEL Classification: G12, G11, D51, D52

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1. Introduction

The notion of arbitrage has been considered as an important conceptual framework for studying the existence of asset market equilibrium since the seminal work Hart (1974). The arbitrage-based literature investigates the existence of equilibrium in asset markets by presupposing that all the preferred sets of investors are unbounded.¹ This presupposition may conflict with satiated preferences because the set of satiations is bounded in general. As discussed below, the mean-variance utility functions may reach satiation in asset markets without riskless assets if they reveal variance aversion. Thus, the existence theorems of the arbitrage-based literature do not apply to the capital asset pricing model (CAPM) without riskless assets. The traditional approaches to equilibrium theory of commodity markets are not applicable to the CAPM with satiation portfolios either, because they exclude the case that satiation occurs only inside the set of feasible and individually rational allocations.

Nielsen (1990) and Allingham (1991) investigate the existence of equilibrium in the classical capital asset pricing model without riskless assets. They assume that agents have homogeneous expectations on the return distribution. Nielsen (1990) focuses on special cases where either risk aversion of agents are constant or satiation portfolios are expressed as multiples of total endowments. Allingham (1991) introduces a technical lemma to show the existence of equilibrium in the case where the mean-variance utility function is strictly concave. The result of Allingham (1991) subsumes as a special case the existence theorem of Nielsen (1990). It does not extend beyond the prototype of the CAPM, however, because the technical lemma of Allingham (1991) does not work any more in the case where investors have heterogeneous expectations on the return distribution.

The purpose of this paper is to show the existence of equilibrium asset prices in the CAPM without riskless assets where agents have heterogeneous expectations on the return distribution and the mean-variance utility functions are quasiconcave.² The

¹For an extensive review of the arbitrage-based approaches, see Dana *et al.* (1999), Page *et al.* (2000), and Allouch (2002) and the references therein.

²For CAPM implications of heterogeneous expectations, see Jarrow (1980) which discusses the effect

mean-variance utility function has distinct features in the absence of riskless assets. More specifically, as shown later, the set of portfolios which are preferred to any given portfolio is bounded under the convexity condition on the preferences. If the utility function is continuous, the preferred set is closed and therefore, compact. This implies that the mean-variance utility function reaches satiation. Equilibrium asset prices may not be positive in the presence of satiation portfolios in general. We present a necessary and sufficient condition under which equilibrium asset prices are strictly positive. Consequently, the result of the paper generalizes the existence proof of Allingham (1991) to the case where agents have heterogeneous expectations on the return distribution and the mean-variance utility functions are quasiconcave.

Sun and Yang (2003) show the existence of equilibrium in markets with heterogeneous expectations by generalizing the approach of Allingham (1991). Won and Chay (2006) examine the existence issue in the CAPM with heterogeneous expectations and concave mean-variance utility functions in the framework of Won and Yannelis (2005). Our approach to the CAPM with heterogeneous expectations has several merits compared to the literature such as Nielsen (1990), Allingham (1991), Sun and Yang (2003), and Won and Chay (2006). First, assumptions which are made here to show the existence of equilibrium and the positivity of equilibrium prices are ‘CAPM-context-free’ so that the approach of the paper can be applicable to asset pricing models beyond the framework of the CAPM. For example, they can be easily adapted to the asset pricing models which take into account the effect of the higher moments of the return distribution such as skewness and kurtosis. This is not the case with the aforementioned literature, however, because their analysis relies on the specific structure of the mean-variance economy. Second, it is not required in this paper that the covariance matrices of the return distribution have full rank. Thus, the approach of the paper can be applied to the markets where redundant assets such as options and futures have their own role in expanding income-transfer opportunities in the presence of portfolio constraints like short-selling restrictions. The existing literature may not be extended to constrained markets with redundant assets, however, because the full-rank condition of heterogeneous expectations on asset prices in the framework of the CAPM.

of the covariance matrices cannot be dispensed with in their framework. Finally, the conditions used here for the existence of equilibrium and the positivity of equilibrium prices are weaker than the literature. Moreover, the paper covers the case where the mean-variance utility functions are quasiconcave while the literature assumes that the mean-variance utility functions are concave.

There exist interesting attempts to address the existence issue with satiable preferences by using a weaker notion of equilibrium. For example, Mas-Colell (1992) introduces ‘equilibrium with slack’ in which allows agents to keep some positive income unused. The result of Mas-Colell (1992) does not apply, however, to asset pricing models because he fails to characterize conditions under which ‘weak equilibrium’ coincides with competitive equilibrium. Allouch and Le Van (2005) extend Mas-Colell (1992) and provide a condition under which the weak equilibrium coincides with Walrasian equilibrium. The consequence of Allouch and Le Van (2005) does not apply to the case where satiation occurs only inside the set of feasible and individually rational allocations.

This paper is organized as follows. In the next section, we introduce an asset market economy and illustrate that equilibrium fails to exist in the presence of satiation portfolios. In section 3, we show that the mean-variance utility function reaches satiation in the absence of riskless portfolios. The main consequence of the paper is provided in Section 4 followed by concluding remarks.

2. Asset Market Equilibrium

There are ℓ assets, indexed by $j = 1, \dots, \ell$. A *portfolio* of assets can be represented by a ℓ -dimensional vector $x \in \mathbb{R}^\ell$, where j th coordinate indicates the number of shares of the j th asset included in the portfolio. There are m investors, indexed by $i = 1, \dots, m$. Let $I = \{1, \dots, m\}$ denote the set of investors. It is assumed that every investor is allowed to take unlimited short sales, i.e., the set of feasible portfolios is $\mathbb{R}^\ell$ for all $i \in I$. Each investor i is endowed with an initial portfolio $e_i \in \mathbb{R}^\ell$ of assets.

For each $j = 1, \dots, \ell$, let $\tilde{r}^j$ denote the return of asset j and $\tilde{r}$ the random vector with $\tilde{r}^j$ as the j th component. Investors are allowed to have heterogeneous expectations about asset returns. Let $\mathbb{E}_i$ denote the expectation operator for investor i . We set $r_i = \mathbb{E}_i[\tilde{r}]$ and $\Omega_i = \mathbb{E}_i[(\tilde{r} - r_i)(\tilde{r} - r_i)^\top]$ where $\top$ denotes the transpose of a matrix. Then r_i and Ω_i denote the mean vector and the covariance matrix of the returns on assets perceived by investor $i \in I$. For a portfolio $x_i \in \mathbb{R}^\ell$, let $\mu_i(x_i)$ and $\sigma_i(x_i)$ denote its mean return and standard deviation, respectively. Clearly, $\mu_i(x_i) = x_i \cdot r_i$ and $\sigma_i(x_i)^2 = x_i^\top \Omega_i x_i$ for all $i \in I$.

We assume that for each $i \in I$, the utility of investor i is represented by a mean-variance function $v_i(\mu, \sigma^2)$ where μ and σ^2 denote the mean and the variance of a portfolio return for investor i . Then the utility of investor i who holds a portfolio $x_i \in \mathbb{R}^\ell$ is expressed as

$$v_i(\mu_i(x_i), \sigma_i(x_i)^2) = v_i(x_i \cdot r_i, x_i^\top \Omega_i x_i).$$

For each $x_i \in \mathbb{R}^\ell$, we set $u_i(x_i) = v_i(\mu_i(x_i), \sigma_i(x_i)^2)$. For a portfolio $x_i \in \mathbb{R}^\ell$, the preferred set $R_i(x_i)$ and the strictly preferred set $P_i(x_i)$ are defined as $R_i(x_i) = \{x'_i \in \mathbb{R}^\ell : u_i(x'_i) \geq u_i(x_i)\}$ and $P_i(x_i) = \{x'_i \in \mathbb{R}^\ell : u_i(x'_i) > u_i(x_i)\}$, respectively. The asset market economy is defined by $\mathcal{E} = (\mathbb{R}^\ell, u_i, e_i)_{i \in I}$.

A collection $x = (x_1, \dots, x_m)$ of portfolios with $x_i \in \mathbb{R}^\ell$ for each $i \in I$ is called an allocation of the economy $\mathcal{E}$. The initial allocation of portfolios is given by $e = (e_1, \dots, e_m)$ and the *market portfolio* by $\sum_{i \in I} e_i$. An *attainable allocation* is an allocation $x \in \mathbb{R}^{\ell m}$ which satisfies $\sum_{i \in I} (x_i - e_i) = 0$. As usual, the budget set of agent i is defined as $\mathcal{B}_i(p) = \{x_i \in \mathbb{R}^\ell : p \cdot x_i \leq p \cdot e_i\}$. An *asset market equilibrium* for the asset market economy $\mathcal{E}$ is a pair $(p, x) \in (\mathbb{R}^\ell \setminus \{0\}) \times \mathbb{R}^{\ell m}$ such that (i) $x_i \in \mathcal{B}_i(p)$ for all $i \in I$, (ii) $P_i(x_i) \cap \mathcal{B}_i(p) = \emptyset$ for all $i \in I$, (iii) $\sum_{i \in I} (x_i - e_i) = 0$. An allocation is said to be individually rational if $x_i \in R_i(e_i)$ for all $i \in I$. We define the set $A = \{x \in \mathbb{R}^{\ell m} : \sum_{i \in I} (x_i - e_i) = 0 \text{ and } x_i \in R_i(e_i) \text{ for all } i \in I\}$. Then the set A is a collection of attainable and individually rational allocations of $\mathcal{E}$. Let A_i be the projection of A onto $\mathbb{R}^\ell$.

To handle the difficulty with satiation, we denote the set of nonsatiated investors at portfolio allocation $x \in A$ by $I(x) = \{i \in I : P_i(x_i) \neq \emptyset\}$ and the set of satiated investors

at portfolio allocation $x \in A$ by $I^s(x) = I \setminus I(x)$. For each $i \in I$, let $S_i = \{x_i \in \mathbb{R}^\ell : P_i(x_i) = \emptyset\}$ denote the set of satiation portfolios. As exemplified below, the presence of satiation portfolios may cause the non-existence of asset market equilibrium.

EXAMPLE 2.1: We take a specific example from the capital asset pricing model with two investors and two assets. Investors are endowed with initial asset holdings $e_1 = (1/2, 1)$ and $e_2 = (3/2, 1)$, respectively. We assume that the two investors have the same expectation about the mean return but different beliefs about the covariance matrix of the asset returns represented by

$$r_1 = r_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \text{and} \quad \Omega_1 = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}, \quad \Omega_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

They have preferences represented by the mean-variance utility function.

$$v_i(\mu, \sigma) = a_i\mu - \frac{1}{2}b_i\sigma^2.$$

We assume that $(a_1, b_1) = (3, 1/2)$ and $(a_2, b_2) = (1, 1)$. Then for a portfolio $x \in \mathbb{R}^2$, the utility function on $\mathbb{R}^2$ has the form

$$\begin{aligned} u_1(x) &= 3x \cdot r_1 - \frac{1}{4}x^\top \Omega_1 x = 3(x_1 + x_2) - \frac{1}{4}(3x_1^2 + 2x_1x_2 + 3x_2^2), \\ u_2(x) &= x \cdot r_2 - \frac{1}{2}x^\top \Omega_2 x = (x_1 + x_2) - \frac{1}{2}(x_1^2 + x_2^2). \end{aligned}$$

It is easy to check that investor 1 is satiated at $s_1 = (3/2, 3/2)$ while investor 2 is satiated at $s_2 = (1, 1)$. As depicted below, indifference curves of agents are an ellipse and a circle centered at the satiation point, respectively. We set $t_1 = (1, 1)$ and $t_2 = (1/2, 1/2)$. Clearly, (s_1, t_2) and (t_1, s_2) are a feasible allocation.

Now we show that this economy has no equilibrium. Suppose to the contrary that (p^*, x^*, y^*) is an equilibrium. There will be three cases to consider: (i) $x^* = s_1$, (ii) $y^* = s_2$, and (iii) $x^* \neq s_1$ and $y^* \neq s_2$. If $x^* = s_1$, then $y^* = t_2$ and therefore $p^* = (1, 1)$.³ In this case, $p^* \cdot s_1 = 3 > 3/2 = p^* \cdot e_1$, which is impossible. If $y^* = s_2$, then $x^* = t_1$ and therefore $p^* = (1, 1)$ by normalization. In this case, $p^* \cdot t_1 = 2 > 3/2 = p \cdot e_1$, which is

³For simplicity, we can take $\lambda = 1$ by normalization.

impossible. Now consider the case (iii). Since $x^* \neq s_1$ and $y^* \neq s_2$, we have $Du_1(x^*) \neq 0$ and $Du_2(y^*) \neq 0$. By the first order condition for utility maximization, there exist $\lambda_1 > 0$ and $\lambda_2 > 0$ such that $p^* = \lambda_1[3(1, 1) - (1/2)(3x_1^* + x_2^*, x_1^* + 3x_2^*)] = \lambda_2[(1, 1) - y^*]$. Since $(x^*, y^*) \in A$, it follows that

$$x^* + y^* = \frac{5}{2}(1, 1) - \left(\frac{1}{4\lambda_1}(3p_1^* + p_2^*) + \frac{1}{\lambda_2}p_1^*, \frac{1}{4\lambda_1}(p_1^* + 3p_2^*) + \frac{1}{\lambda_2}p_2^* \right) = (2, 2).$$

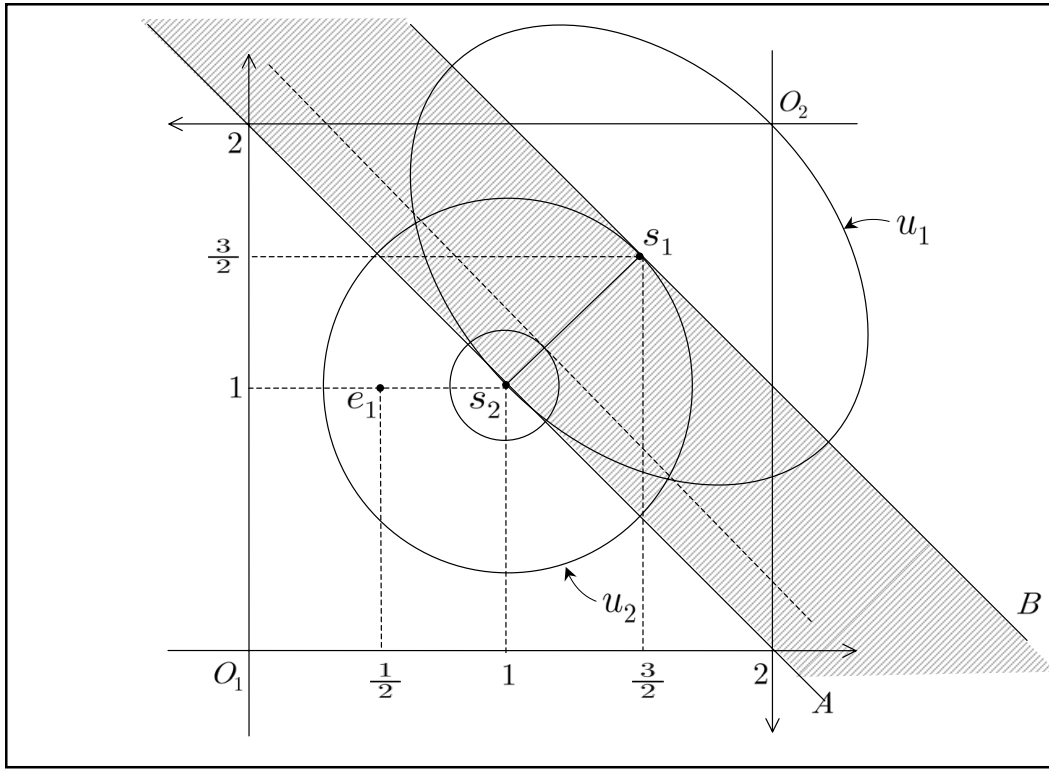


FIGURE 1 : Non-Existence of Equilibrium

Due to symmetry, one can show that $p_1^* = p_2^*$ and therefore $x_1^* = x_2^*$ and $y_1^* = y_2^*$. Putting these outcomes in the budget constraints, we see that $x^* = (3/4, 3/4)$ and $y^* = (5/4, 5/4)$. But these results are contradictory because $p^* = \lambda_1[3(1, 1) - (1/2)(3x_1^* + x_2^*, x_1^* + 3x_2^*)] = \lambda_1(3/2, 3/2)$, while $p^* = \lambda_2[(1, 1) - y^*] = -\lambda_2(1/4, 1/4)$. Hence this economy has no equilibrium.

The non-existence of equilibrium can be checked in the above diagram. The line segment between s_1 and s_2 corresponds to the set of Pareto optimal allocations. The price which supports an optimal allocation is a vector in the line through s_1 and s_2 . Let A and B denote a line orthogonal to the supporting price at an allocation where one of the agents is satiated. Notice that the initial allocation $e = (e_1, e_2)$ is outside the closed band between the lines A and B . Thus, any budget line which passes through the initial endowment and an optimal allocation fails to be orthogonal to the line through s_1 and s_2 . $\square$

3. Characterizations of the CAPM

We characterize the properties of asset markets where agents are allowed to have heterogeneous expectations on the return distribution and the mean-variance utility functions are quasiconcave. To do this, for each $i \in I$, we make the following assumptions:

C1. v_i is differentiable and strictly quasiconcave.⁴

C2. For each point $(\mu, \sigma^2) \in \mathbb{R}^2$ and any positive number a ,

$$v_i(\mu, \sigma^2) \leq v_i(\mu + a, \sigma^2), \quad v_i(\mu, \sigma^2 + a) < v_i(\mu, \sigma^2).$$

C3. Each Ω_i is positive definite.

Assumption **C1** subsumes the conditions of Allingham (1991) on preferences as a special case because Allingham (1991) assumes that v_i is strictly concave. Assumption **C2** states that v_i is increasing in expected return and strictly decreasing in variance. Assumption **C3** implies the absence of nontrivial riskless portfolios. The following lemma shows that u_i is differentiable and quasiconcave.

⁴A function $f : \mathbb{R}^\ell \rightarrow \mathbb{R}$ is strictly quasiconcave if for all $\lambda \in (0, 1)$ and x, x' in $\mathbb{R}^\ell$, $f(x') > f(x)$ implies $f(\lambda x + (1 - \lambda)x') > f(x)$.

LEMMA 3.1: The utility function u_i is differentiable and strictly quasiconcave under Assumptions **C1-C3**.

PROOF : By Assumption **C1**, v_i is differentiable. Since μ_i and σ_i are a differentiable function from $\mathbb{R}^\ell$ to $\mathbb{R}$, so is u_i as a composite of differentiable functions. By Assumption **C3**, the quadratic form $x^\top \Omega_i x$ is a convex function of x . Let x_i and z_i be a point in $\mathbb{R}^\ell$. Then it follows that for all α in $[0, 1]$,

$$(\alpha x_i + (1 - \alpha)z_i)^\top \Omega_i (\alpha x_i + (1 - \alpha)z_i) \leq \alpha x_i^\top \Omega_i x_i + (1 - \alpha)z_i^\top \Omega_i z_i.$$

Since v_i is strictly quasiconcave and nonincreasing in the second argument, we see that for all α in $[0, 1]$,

$$\begin{aligned} u_i(\alpha x_i + (1 - \alpha)z_i) &= v_i(\alpha x_i \cdot r_i + (1 - \alpha)z_i \cdot r_i, (\alpha x_i + (1 - \alpha)z_i)^\top \Omega_i (\alpha x_i + (1 - \alpha)z_i)) \\ &\geq v_i(\alpha x_i \cdot r_i + (1 - \alpha)z_i \cdot r_i, \alpha x_i^\top \Omega_i x_i + (1 - \alpha)z_i^\top \Omega_i z_i) \\ &> \min\{v_i(x_i \cdot r_i, x_i^\top \Omega_i x_i), v_i(z_i \cdot r_i, z_i^\top \Omega_i z_i)\} \\ &= \min\{u_i(x_i), u_i(z_i)\}. \end{aligned}$$

■

By Lemma 3.1, $P_i(x_i)$ is a convex set for all $x_i \in \mathbb{R}^\ell$. The following lemma shows that the set $R_i(x_i)$ is compact for all $x_i \in \mathbb{R}^\ell$.

LEMMA 3.2: Suppose that **C1-C3** hold. Then for each $x_i \in \mathbb{R}^\ell$, $R_i(x_i)$ is compact.

PROOF : By Assumption **C1** and Lemma 3.1, $R_i(x_i)$ is closed and convex for any $x_i \in \mathbb{R}^\ell$. Suppose that $R_i(x_i)$ is not bounded. Then there exists a nonzero $y \in \mathbb{R}^\ell$ which is a direction of recession of $R_i(x_i)$.⁵ For any $\lambda \geq 0$, therefore, we have $u_i(x_i + \lambda y) \geq u_i(x_i)$.

⁵A vector v is said to be a direction of recession of a set S in $\mathbb{R}^\ell$ if there exists a point $x \in S$ such that $x + \lambda v \in S$ for every $\lambda \in \mathbb{R}_+$.

It follows from the quasiconcavity of v_i that, for any $\alpha \in [0, 1]$,

$$\begin{aligned}
& v_i((1 - \alpha)x_i \cdot r_i + \alpha\lambda(x_i/\lambda + y) \cdot r_i, (1 - \alpha)x_i^\top \Omega_i x_i + \alpha\lambda^2(x_i/\lambda + y)^\top \Omega_i(x_i/\lambda + y)) \\
&= v_i((1 - \alpha)x_i \cdot r_i + \alpha(x_i + \lambda y) \cdot r_i, (1 - \alpha)x_i^\top \Omega_i x_i + \alpha(x_i + \lambda y)^\top \Omega_i(x_i + \lambda y)) \\
&> \min\{v_i(x_i \cdot r_i, x_i^\top \Omega_i x_i), v_i((x_i + \lambda y) \cdot r_i, (x_i + \lambda y)^\top \Omega_i(x_i + \lambda y))\} \\
&= \min\{u_i(x_i), u_i(x_i + \lambda y)\} \\
&= u_i(x_i).
\end{aligned}$$

By setting $\alpha = 1/\lambda^2$ and $\lambda \rightarrow \infty$, the previous relations yield

$$v_i(x_i \cdot r_i, x_i^\top \Omega_i x_i + y^\top \Omega_i y) \geq v_i(x_i \cdot r_i, x_i^\top \Omega_i x_i).$$

Since Assumption **C3** implies $y^\top \Omega_i y > 0$, the above inequality contradicts Assumption **C2**. ■

For each $\gamma \in \mathbb{R}$ and each $i \in I$, we set $x_i(\gamma) = \gamma \Omega_i^{-1} r_i$ and $\theta_i = r_i^\top \Omega_i^{-1} r_i$. The results of Lemma 3.2 leads to the following proposition.

PROPOSITION 3.1: Suppose that Assumptions **C1-C3** is satisfied. Then for each $i \in I$, the following hold:

- (i) The set $R_i(e_i)$ is compact.
- (ii) There exist satiation portfolios in $R_i(e_i)$, i.e., $S_i \neq \emptyset$.
- (iii) Suppose that each u_i is differentiable and $r_i \neq 0$ for all $i \in I$. Then for some $\gamma_i \in \mathbb{R}$, $x_i(\gamma_i) = \gamma_i \Omega_i^{-1} r_i$ is a satiation portfolio if and only if γ_i solves the following equation

$$D_1 v_i(\gamma_i \theta_i, \gamma_i^2 \theta_i) + 2\gamma_i D_2 v_i(\gamma_i \theta_i, \gamma_i^2 \theta_i) = 0,$$

where, for each $j = 1, 2$, D_j denotes the partial differentiation with respect to the j -th argument of v_i .

PROOF : (i) By Lemma 3.2, $R_i(e_i)$ is compact. □

(ii) Since u_i is continuous and $R_i(e_i)$ is compact, there must exist a portfolio in $R_i(e_i)$ which maximizes u_i over $\mathbb{R}^\ell$. $\square$

(iii) Let x_i denote a satiation portfolio for each $i \in I$. Then the first-order condition for utility maximization yields

$$D_1 v_i(x_i \cdot r_i, x_i^\top \Omega_i x_i) r_i + 2D_2 v_i(x_i \cdot r_i, x_i^\top \Omega_i x_i) \Omega_i x_i = 0. \quad (1)$$

We set

$$\gamma_i = -\frac{D_1 v_i(x_i \cdot r_i, x_i^\top \Omega_i x_i)}{2D_2 v_i(x_i \cdot r_i, x_i^\top \Omega_i x_i)}.$$

Then (1) gives $x_i = \gamma_i \Omega_i^{-1} r_i$. By putting x_i into (1), we have

$$[D_1 v_i(\gamma_i \theta_i, \gamma_i^2 \theta_i) + 2\gamma_i D_2 v_i(\gamma_i \theta_i, \gamma_i^2 \theta_i)] r_i = 0. \quad (2)$$

Since $r_i \neq 0$, it follows that

$$D_1 v_i(\gamma_i \theta_i, \gamma_i^2 \theta_i) + 2\gamma_i D_2 v_i(\gamma_i \theta_i, \gamma_i^2 \theta_i) = 0. \quad (3)$$

Suppose that there exists γ_i which solves (3). We set $x_i = \gamma_i \Omega_i^{-1} r_i$. By putting it into (2), we obtain the same relation as in (1). This implies that x_i is a satiation point of u_i . $\blacksquare$

4. Existence of Positive Equilibrium Prices

In this section, we show the existence of equilibrium and the positivity of equilibrium prices in the CAPM. The condition used here for the existence of equilibrium is shown to be weaker than the assumptions used in the literature.

We define the function $b_i : \mathbb{R}^\ell \rightarrow \mathbb{R}$ such that for all $x_i \in \mathbb{R}^\ell$,

$$b_i(x_i) = -\frac{D_2 v_i(x_i \cdot r_i, x_i^\top \Omega_i x_i)}{D_1 v_i(x_i \cdot r_i, x_i^\top \Omega_i x_i)}.$$

The function b_i measures the degree of risk aversion in the CAPM.⁶ For each $x \in \mathbb{R}^{\ell m}$ with $I(x) \neq \emptyset$, we define the set

$$H(x) = \sum_{i \in I(x)} \text{con} [P_i(x_i) - \{x_i\}]$$

where, for a set $S \subset \mathbb{R}^\ell$, $\text{con}(S) = \bigcup_{\lambda > 0} \lambda S$, i.e., $\text{con}(S)$ is the cone generated by the set S . Since $P_i(x_i)$ is open and convex, $H(x)$ is an open, convex cone. For each $i \in I$, we make the following assumptions.

D1. Let x be an allocation in A with $I^s(x) \neq \emptyset$. Then for each $k \in I^s(x)$, there exists $i \in I(x)$ such that

$$\nabla u_i(x_i) \cdot (x_k - e_k) \geq 0.$$

D2. For all $x \in A$, $\mathbb{R}_+^\ell \setminus \{0\} \subset H(x)$.

Assumption **D1** requires that for all $x \in A$ with $I^s(x) \neq \emptyset$, there be an agent $i \in I(x)$ for each $k \in I^s(x)$ which evaluates the satiation portfolio x_k of agent k more than the initial endowment e_k with the marginal valuation at his own choice x_i . By exploiting the special property of mean-variance utility functions, Assumption **D1** is written in an alternative way:

D1a. Let x be an allocation in A with $I^s(x) \neq \emptyset$. Then for each $k \in I^s(x)$, there exists $i \in I(x)$ such that

$$\left(\frac{r_i}{2b_i(x_i)} - \Omega_i x_i \right) \cdot (x_k - e_k) \geq 0.$$

Assumption **D1** alone is required for the existence of equilibrium. Assumption **D2** ensures the strict positivity of equilibrium prices. As shown below, this condition is necessary and sufficient for equilibrium prices to be strictly positive.

The following example shows that Assumption **D1** is necessary and sufficient for equilibrium to exist in the economy where agents have the same preferences as in Example 2.1.

⁶For details, see Allingham (1991).

EXAMPLE 4.1: In the economy of Example 2.1, we have had $t_1 = s_2 = (1, 1)$. We set $x = (t_1, s_2)$. Then x is a feasible allocation and $I(x) = \{1\}$. Recall that for some $\lambda > 0$, $\nabla u_1(t_1) = \lambda(1, 1)$. Thus, we have $(1, 1) \cdot (s_2 - e_2) = -1/2 < 0$ and therefore, $\nabla u_1(t_1) \cdot (s_2 - e_2) < 0$. Thus, Assumption **D1** is violated in the economy of Example 2.1.

In fact, we can show that **D1** is necessary and sufficient for equilibrium to exist in the economy which is the same as in Example 2.1 except that $e_1 = (a, b)$ and $e_2 = (2 - a, 2 - b)$ for some $(a, b) \in \mathbb{R}^2$. To check **D1** in the modified economy, we consider the inequalities $(1, 1) \cdot (s_1 - e_1) = (1, 1) \cdot [(3/2, 3/2) - (a, b)] \geq 0$ and $(1, 1) \cdot (s_2 - e_2) = (1, 1) \cdot [(1, 1) - (2 - a, 2 - b)] \geq 0$. Both inequalities yield $2 \leq a + b \leq 3$. It is easy to check in Figure 1 that equilibrium exists in the modified economy of Example 2.1 if and only if $2 \leq a + b \leq 3$. The region of the endowments of agent 1 which satisfy Assumption **D1** is depicted by the shaded area between lines A and B in Figure 1. $\square$

Two remarks on the assumptions **D1** and **D2** are in order.

REMARK 1: The conditions **D1** and **D2** are quite different from those of Sun and Yang (2003) and moreover, strictly weaker than them, as shown below. For the purpose of comparison, we reproduce ‘Assumption 1’ of Sun and Yang (2003) which is made to build their main existence theorem:

SY. For every nonzero $\alpha \in \mathbb{R}_+^m$, every $i \in I$, and every $\beta \in V$, it holds that

$$(i) \sum_{i \in I} \left(\frac{\Omega_i^{-1} r_i}{2\beta_i} - e_i \right) \neq 0 \quad \text{and}$$

$$(ii) \left(\frac{\Omega_i^{-1} r_i}{2\beta_i} - e_i \right)^\top \left(\sum_{i \in I} \alpha_i \Omega_i^{-1} \right)^{-1} \left[\sum_{i \in I} \left(\frac{\Omega_i^{-1} r_i}{2\beta_i} - e_i \right) \right] \geq 0,$$

where $V = \{(\beta_1, \dots, \beta_m) \in \mathbb{R}^m : \beta_i = b_i(x_i), i \in I, \text{ for some } x_i \text{ with } u_i(x_i) \geq u_i(e_i)\}$.

CLAIM: The condition **SY** implies Assumption **D1**.

PROOF : For each $i \in I$, let $\bar{x}_i$ be a satiation portfolio, i.e., $\bar{x}_i \in S_i$. By (iii) of Proposition

3.1, we have $\bar{x}_i = \Omega_i^{-1} r_i / 2b_i(\bar{x}_i)$ for each $i \in I$. Suppose that the economy satisfies the condition **SY**. In particular, (i) of the condition **SY** implies that $\sum_{i \in I} (\bar{x}_i - e_i) \neq 0$ or $I(x) \neq \emptyset$ for all $x \in A$. Let x be an allocation in A . Recalling that $\Omega_i^{-1} r_i / 2b_i(x_i) - x_i = 0$ for each $i \in I^s(x)$, it follows from (ii) of the condition **SY** that for each nonzero $\alpha \in \mathbb{R}_+^m$, each $x \in X$, and each $k \in I^s(x)$,

$$\begin{aligned} 0 &\leq \left(\frac{\Omega_k^{-1} r_k}{2b_k(x_k)} - e_k \right)^\top \left(\sum_{i \in I} \alpha_i \Omega_i^{-1} \right)^{-1} \left[\sum_{i \in I} \left(\frac{\Omega_i^{-1} r_i}{2b_i(x_i)} - e_i \right) \right] \\ &= (x_k - e_k)^\top \left(\sum_{i \in I} \alpha_i \Omega_i^{-1} \right)^{-1} \left[\sum_{i \in I} \left(\frac{\Omega_i^{-1} r_i}{2b_i(x_i)} - x_i \right) \right] \\ &= (x_k - e_k)^\top \left(\sum_{i \in I} \alpha_i \Omega_i^{-1} \right)^{-1} \left[\sum_{i \in I(x)} \left(\frac{\Omega_i^{-1} r_i}{2b_i(x_i)} - x_i \right) \right]. \end{aligned}$$

Thus, for each $k \in I^s(x)$, there exists $i \in I(x)$ such that

$$(x_k - e_k)^\top \left(\sum_{i \in I} \alpha_i \Omega_i^{-1} \right)^{-1} \left(\frac{\Omega_i^{-1} r_i}{2b_i(x_i)} - x_i \right) \geq 0. \quad (4)$$

Now we choose α in $\mathbb{R}_+^m$ such that $\alpha_i = 1$ and $\alpha_j = 0$ for all $j \neq i$. Then it follows from (4) that

$$(x_k - e_k) \cdot \left(\frac{r_i}{2b_i(x_i)} - \Omega_i x_i \right) \geq 0.$$

Thus, we conclude that the condition **SY** implies Assumption D1. $\square$

Readers will recognize that Assumption **D1** is strictly weaker (in fact, much weaker) than the condition **SY** when they read through the proof of CLAIM. The difference between them is also found in markets where utility functions do not reach satiation. In this case, Assumption **D1** trivially holds but the condition **SY** imposes certain restrictions on preferences.

REMARK 2: Assumption **D1** has an additional advantage over the conditions of Sun and Yang (2003) and Allingham (1991). They assume that the covariance matrix Ω_i has full rank for each $i \in I$ by allowing no redundant assets like options and futures

in markets. This assumption is not valid any more in asset markets where portfolio constraints such as short-selling restrictions enables redundant assets to create new opportunities for income transfers. In contrast, the full rank condition of the covariance matrix is not required to build the main consequences of the paper. Thus, our approach can be extended to constrained asset markets with redundant assets.

The following result shows that **D2** is necessary and sufficient for equilibrium prices to be strictly positive.

PROPOSITION 4.1: Suppose that u_i is differentiable for all $i \in I$. Let (p, x) be an equilibrium with $I(x) \neq \emptyset$. Then **D2** holds if and if $p \gg 0$.

PROOF : Since (p, x) be an equilibrium, for each $i \in I(x)$, $y_i \in P_i(x_i)$ implies that $p \cdot x_i < p \cdot y_i$ and therefore, $p \cdot z_i > 0$ for all $z_i \in \text{con}[P_i(x_i) - \{x_i\}]$. Since u_i is differentiable, p is the unique vector which supports $\text{con}[P_i(x_i) - \{x_i\}]$ at zero and therefore, $\text{con}[P_i(x_i) - \{x_i\}]$ equals the open half space $\{z \in \mathbb{R}^\ell : p \cdot z > 0\}$. Thus we have $H(x) = \{z \in \mathbb{R}^\ell : p \cdot z > 0\}$.

Suppose that $p \gg 0$. Then $p \cdot z > 0$ for all $z \in \mathbb{R}_+^\ell \setminus \{0\}$. This implies $\mathbb{R}_+^\ell \setminus \{0\} \subset H(x)$. Conversely, suppose that $\mathbb{R}_+^\ell \setminus \{0\} \subset H(x)$. For each $j \in \{1, \dots, \ell\}$, let $\mathbf{1}_j$ denote the vector in $\mathbb{R}^\ell$ which has 1 in its j th coordinate and zero in the other coordinates. Clearly, $\mathbf{1}_j \in \mathbb{R}_+^\ell \setminus \{0\}$. Since $\mathbb{R}_+^\ell \setminus \{0\} \subset H(x)$, we must have $p \cdot \mathbf{1}_j = p_j > 0$ for all $j \in \{1, \dots, \ell\}$ where p_j denotes the j th coordinate of p . Therefore, we conclude that $p \gg 0$. ■

We provide the existence of equilibrium in the CAPM with satiable preferences and heterogeneous expectations on the mean and variance of the return distribution. This result extends the equilibrium existence theorem of Allingham's (1991) homogeneous expectation model to the case with heterogeneous expectation and moreover, generalizes the existence theorem of Sun and Yang (2003).

THEOREM 4.1: Suppose that $\sum_{i \in I} e_i \notin \sum_{i \in I} S_i$.⁷ Then under Assumptions **C1-C3**

⁷The supposition excludes the unrealistic case that every agent reaches satiation in equilibrium. For details on the existence of equilibrium in this case, see Won and Yannelis (2005).

and **D1-D2**, the CAPM has an equilibrium (p, x) with $p \gg 0$.

PROOF : Let Δ be the unit closed ball in $\mathbb{R}^\ell$. According to (i) of Proposition 3.1, A is bounded, so that one can find a compact convex cube K with center 0 such that $A_i \cup \Delta \subset \text{int}K$ for each $i \in I$. For each $i \in I$, we set $X_i = K$. Then it is obvious that X_i is compact and convex and $A_i \subset X_i$ for every $i \in I$. We set $X = \prod_{i \in I} X_i$ and define the correspondences $\hat{\mathcal{B}}_i : \Delta \rightarrow 2^{X_i}$ and $\hat{\mathcal{B}}_i^\circ : \Delta \rightarrow 2^{X_i}$ such that for each $p \in \Delta$,

$$\begin{aligned}\hat{\mathcal{B}}_i(p) &= \{x_i \in X_i : p \cdot x_i \leq p \cdot e_i + (1 - \|p\|)\}, \\ \hat{\mathcal{B}}_i^\circ(p) &= \{x_i \in X_i : p \cdot x_i < p \cdot e_i + (1 - \|p\|)\}.\end{aligned}$$

Notice that $e_i \in A_i \subset \text{int}K = \text{int}X_i$ and therefore, $\hat{\mathcal{B}}_i^\circ(p) \neq \emptyset$ for all $p \in \Delta$.

We set $X_0 = \Delta$ and $I_0 = I \cup \{0\}$, and consider the abstract economy $\Gamma = (X_i, \mathcal{A}_i, \mathcal{P}_i)_{i \in I_0}$ where

- (1) the correspondence $\mathcal{A}_i : \Delta \times X \rightarrow 2^{X_i}$ for each $i \in I_0$ is defined such that $\mathcal{A}_0(p, x) = \Delta$ and $\mathcal{A}_i(p, x) = \hat{\mathcal{B}}_i^\circ(p)$ for each $i \in I$, and
- (2) the correspondence $\mathcal{P}_i : \Delta \times X \rightarrow 2^{X_i}$ for each $i \in I_0$ is defined such that

$$\begin{aligned}\mathcal{P}_0(p, x) &= \{q \in \Delta : q \cdot \sum_{i \in I} (x_i - e_i) > p \cdot \sum_{i \in I} (x_i - e_i)\}, \\ \mathcal{P}_i(p, x) &= P_i(x_i) \cap K := \hat{P}_i(x_i), \forall i \in I.\end{aligned}$$

Since $\mathcal{E}$ satisfies Assumptions **C1-C3** for all $i \in I$, the abstract economy Γ satisfies the conditions of Theorem 6.1 of Yannelis and Prabhakar (1983). Thus, there must exist a pair $(p^*, x^*) \in \Delta \times X$ which satisfies (a) $x_i^* \in \hat{\mathcal{B}}_i(p^*)$, $\forall i \in I$, (b) $\hat{P}_i(x_i^*) \cap \hat{\mathcal{B}}_i^\circ(p^*) = \emptyset$, $\forall i \in I$, and (c) $p^* \cdot z^* \geq p \cdot z^*$, $\forall p \in \Delta$, where $z_i^* = x_i^* - e_i$ and $z^* = \sum_{i \in I} (x_i^* - e_i)$.

To see that $z^* = 0$, suppose to the contrary that $z^* \neq 0$. Then it follows from (c) that $p^* \cdot z^* > 0$ and $\|p^*\| = 1$. Then by (a), we obtain $p^* \cdot z^* \leq 0$, which leads to a contradiction. Hence, we have $z^* = 0$. Since $\sum_{i \in I} e_i \notin \sum_{i \in I} S_i$, we have $(S_1 \times \cdots \times S_n) \cap A = \emptyset$ and therefore, $I(x) \neq \emptyset$ for all $x \in A$. Thus, we obtain $I(x^*) \neq \emptyset$.

We claim that $p^* \cdot x_i^* = p^* \cdot e_i + (1 - \|p^*\|)$, $\forall i \in I(x^*)$. Since $x_i^* \in \hat{\mathcal{B}}_i(p^*)$, we have $p^* \cdot x_i^* \leq p^* \cdot e_i + (1 - \|p^*\|)$. We need to show $p^* \cdot x_i^* \geq p^* \cdot e_i + (1 - \|p^*\|)$. Since

$i \in I(x^*)$ and $x_i^* \in \text{int}K$, $\hat{P}_i(x_i^*) \neq \emptyset$. Thus, we can choose $x_i \in \hat{P}_i(x_i^*)$. Then it follows from (b) that $\alpha x_i + (1 - \alpha)x_i^* \in \hat{P}_i(x_i^*)$ for any $\alpha \in (0, 1]$ and $p^* \cdot [\alpha x_i + (1 - \alpha)x_i^*] \geq p^* \cdot e_i + (1 - \|p^*\|)$. As $\alpha \rightarrow 0$, we have $p^* \cdot x_i^* \geq p^* \cdot e_i + (1 - \|p^*\|)$. This proves that $p^* \cdot x_i^* = p^* \cdot e_i + (1 - \|p^*\|)$, $\forall i \in I(x^*)$. Observe that for each $i \in I(x^*)$, $x'_i \in P_i(x_i^*)$ has $p^* \cdot x'_i \geq p^* \cdot e_i + (1 - \|p\|) = p \cdot x_i^*$. That is, p^* supports the convex set $P_i(x_i^*)$ at x_i^* . Since u_i is differentiable, this implies that there exists $\lambda_i > 0$ for each $i \in I(x^*)$ such that $\lambda_i p^* = \nabla u_i(x_i^*)$.

We show that $\|p^*\| = 1$ and $p^* \cdot x_i^* = p^* \cdot e_i$, $\forall i \in I$. Suppose that $\|p^*\| < 1$. Then $p^* \cdot x_i^* = p^* \cdot e_i + (1 - \|p^*\|) > p^* \cdot e_i$ for every $i \in I(x^*)$. On the other hand, by Assumption **D1**, there exists $i \in I(x^*)$ for each $k \in I^s(x^*)$ such that $\nabla u_i(x_i^*) \cdot (x_k^* - e_k) \geq 0$. Thus, we have $p^* \cdot (x_k^* - e_k) \geq 0$ for all $k \in I^s(x^*)$. Summing up these inequalities over I , we obtain $p^* \cdot z^* > 0$, which contradicts $z^* = 0$. Thus, it must be the case that $\|p^*\| = 1$ and $p^* \cdot x_i^* = p^* \cdot e_i$, $\forall i \in I$.

Consequently, it holds that $x_i^* \in \mathcal{B}_i(p^*)$ and $\hat{P}_i(x^*) \cap \{x_i \in X_i : p^* \cdot x_i < p^* \cdot e_i\} = \emptyset$, $\forall i \in I$. Since x_i^* is in the interior of K for all $i \in I$, it is easy to see that $P_i(x^*) \cap \mathcal{B}_i^\circ(p^*) = \emptyset$ where $\mathcal{B}_i^\circ(p^*) = \{x_i \in \mathbb{R}^\ell : p^* \cdot x_i < p^* \cdot e_i\}$, $\forall i \in I$. Since both $P_i(x^*)$ and $\mathcal{B}_i^\circ(p^*)$ are open and $\mathcal{B}_i^\circ(p^*) \neq \emptyset$ for every $i \in I$, this implies that $P_i(x^*) \cap \mathcal{B}_i(p^*) = \emptyset$, $\forall i \in I$. Hence, (p^*, x^*) constitutes an equilibrium of the asset market economy $\mathcal{E}$. It follows by Proposition 4.1 that $p^* \gg 0$. ■

5. Conclusion

This paper shows the existence of equilibrium and the positivity of equilibrium prices in the mean-variance economy with heterogeneous expectations on the return distribution. These consequences are based upon new conditions on the utility functions and the initial allocation of portfolios.

In contrast to the literature, our approach to the existence of equilibrium prices in the CAPM does not rely on the structure of the mean-variance economy. This is a definite advantage over the approach of Allingham (1991) and Sun and Yang (2003)

because the current approach enables us to analyze asset pricing models beyond the framework of the traditional CAPM. Specifically, the consequences of the paper can be generalized to the case where higher-moments of the return distribution like skewness and kurtosis matter to asset pricing. The other possible extension is to investigate the existence of equilibrium asset prices with redundant assets such as options and futures in asset markets which are subject to portfolio constraints.

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